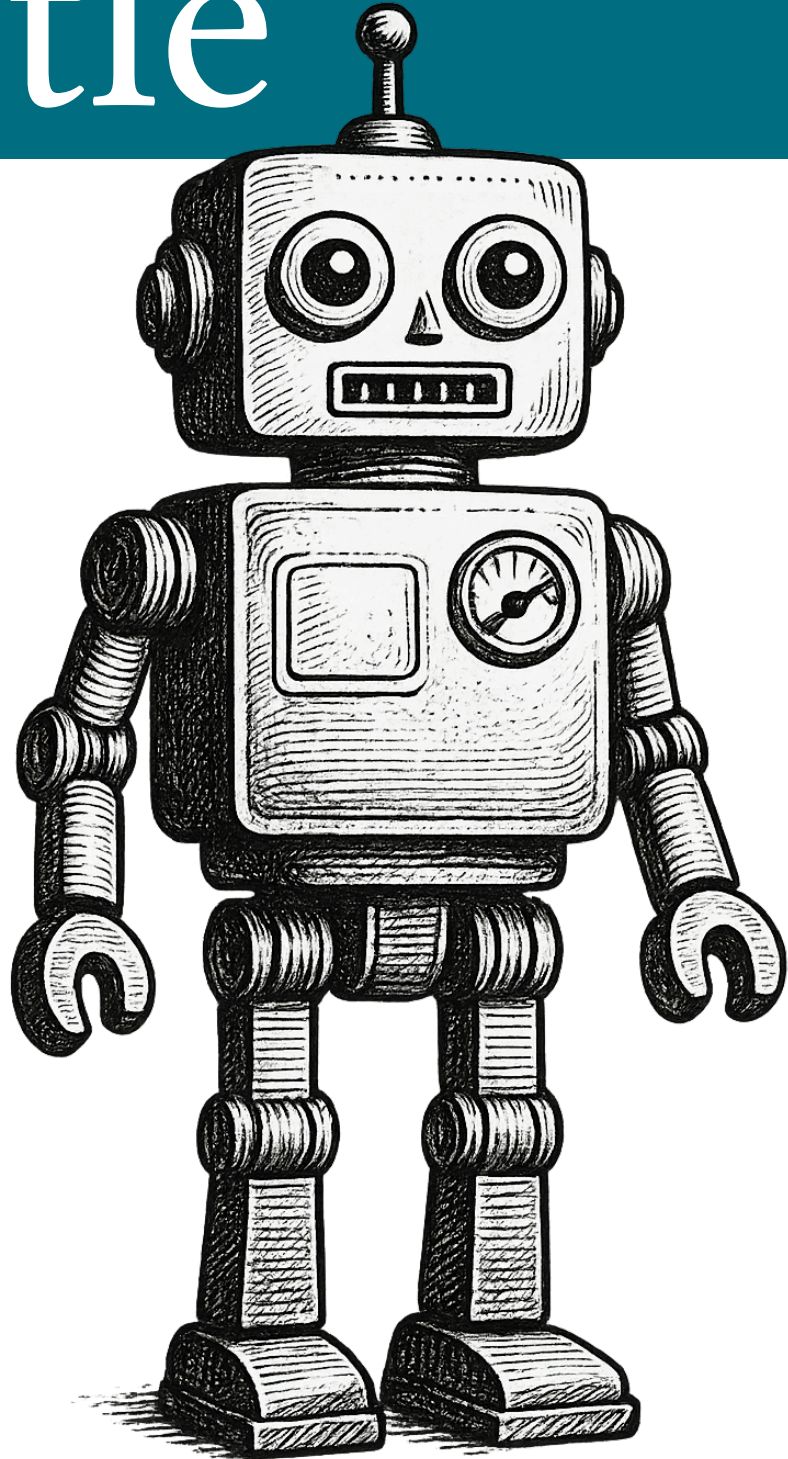


Botnet in a

Bottle



bottle

A novel approach to large-scale botnet analysis

by

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to obtain the degree of Master of Science

at the Delft University of Technology,

to be defended publicly on Wednesday December 17, 2025 at 10:00

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Project duration: September 2025 – May 2026

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Preface

This is a preface where I sound incredibly grateful and inspired.

Abstract

Here is were I am surprisingly concrete for a section that is called abstract.

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Introduction

Availability is a foundational property of online services. Modern societies increasingly rely on digital infrastructure under the assumption that critical services will be reachable when needed. Recent and not-so-recent incidents, such as the Ukraine power grid blackout, demonstrate how disruptions of online availability can lead to significant societal and economic damage [1]. As a result, effective mitigation of Denial of Service (DoS) attacks remains a central challenge in cybersecurity. Addressing this challenge requires more than traffic filtering alone; it demands an understanding of the actors, incentives, and organizational structures behind such attacks.

A clear illustration of this need can be found in the DDoS activity attributed to the “Gorilla” ecosystem, which surged in September 2024 [2]. During this period, a Swiss national critical-infrastructure operator experienced overload attacks that temporarily disrupted the availability of its online services. Subsequent analysis by the Swiss National Cyber Security Centre (NCSC) linked these incidents to a DDoS-as-a-service operation advertised via Telegram under names such as “Gorilla Services.” The attacks were executed using a botnet (a network of compromised machines remotely controlled by an operator) combined with rented, high-bandwidth infrastructure capable of generating large volumes of traffic [3].

In this ecosystem, infected devices run malware that communicates with Command and Control (C2) servers, which distribute attack instructions and coordinate activity across the botnet. Beyond this technical infrastructure, Gorilla actively markets its services and maintains payment and communication channels, enabling it to rapidly recover from platform takedowns, such as the removal and replacement of its Telegram channels reported by the NCSC [2]. This organizational complexity demonstrates that contemporary DDoS attacks are not merely technical events, but coordinated socio-technical operations. Consequently, effective defensive measures depend on actionable intelligence about both the underlying botnet infrastructure and the actors who operate it.

One approach to obtaining such intelligence is to study botnet activity at a high level, including which botnets are active, which targets they select, and which techniques they use to execute attacks. However, comprehensive and publicly available overviews of botnet activity are currently lacking. This is due to several factors, including legal and ethical constraints, as well as the technical difficulty of collecting such intelligence at scale.

A common method for extracting botnet activity is to analyze the Command and Control (C2) server that distributes instructions to infected hosts. This typically involves reverse-engineering the malware binary used to enroll a host into a botnet [citation]. By understanding how a bot communicates with its C2 server, it is possible to emulate this communication and collect information without directly participating in attacks. Applications that implement this approach are commonly referred to as C2 miners [citation].

While effective, C2 mining through reverse-engineering does not scale easily, as each botnet family requires significant manual effort and specialized expertise. As a result, maintaining a comprehensive and timely overview of botnet activity remains a largely manual and fragmented process. Addressing this limitation requires an alternative approach to large-scale botnet intelligence gathering.

1.1 Botnet in a Bottle

In addressing the challenge of scalable botnet intelligence gathering, this thesis proposes *bottle*, an approach that observes the behavior of botnet malware in a controlled environment rather than relying on detailed reverse-engineering of individual botnet implementations. At a high level, the idea is to allow malware to communicate with its command infrastructure while preventing harm. By executing malware without modification, this approach reduces the amount of manual, botnet-specific effort required to extract intelligence from newly emerging malware families.

This shift from botnet-specific reverse-engineering to behavioral monitoring aims to enable the scalable collection of technical intelligence about active botnets and their operational characteristics, while remaining agnostic to individual malware implementations.

1.2 Thesis Structure

This thesis is organized into three parts. Part 1 provides a general introduction and outlines the high-level problem addressed in this work, as well as the core idea proposed to address it. It is intended to be accessible to readers without a background in cybersecurity or malware analysis.

Part 2 presents the technical background required to understand the proposed approach. This includes an introduction to botnets (Section 2.1), how intelligence about botnet activity can be used defensively (Section 2.2), and the tools and techniques used to observe malware behavior in controlled environments (Section 2.3 and Section 2.4).

Part 3 consists of a scientific article that presents the proposed approach and its evaluation. The article builds directly on the background introduced in Part 2 and focuses on the design, implementation, and analysis of the system proposed in this thesis.

Background

This part of the thesis provides the background and terminology required to make the scientific article presented in Chapter 3 accessible to non-expert readers. It introduces the fundamental concepts needed to understand both the problem domain and the proposed approach.

Section 2.1 and Section 2.2 describe the basic operation of botnets and explain how intelligence about their activity can be used to improve cyber defences. Then, Section 2.3 and Section 2.4 cover the technical foundations necessary to understand the implementation discussed in the scientific paper.

2.1 How Botnets Work

2.2 Intelligence for Defenders

2.3 Malware Sandboxes

2.4 Network Traffic Analysis

References

- [1] J. Finkle, U.S. Firm Blames Russian 'Sandworm' Hackers for Ukraine Outage, Reuters (2016).
- [2] F. D. o. D. DDPS Civil Protection and Sport, Brief Technical Analysis of the "Gorilla" Botnet, (n.d.).
- [3] M. Weyns, Exploring the Gorillas in the Malware Jungle, (2025).

